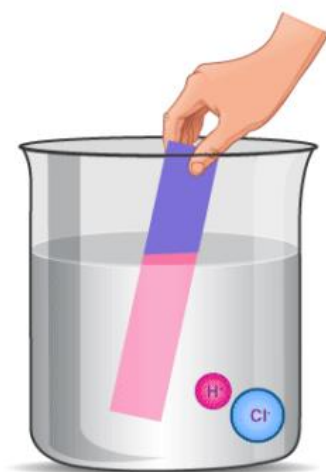


**KWA-THEMA SKILL SCHOOL**

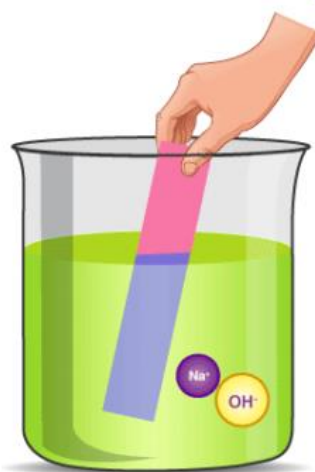
**NS-TECH WORK BOOK**

**YEAR 4 TERM 2**



**Acid**

Blue litmus turns red



**Base**

Red litmus turns blue

© Byjus.com

*teachoo*

# DIFFERENCE BETWEEN ACID AND BASES



**Acid**



**Base**

**NAME:** \_\_\_\_\_

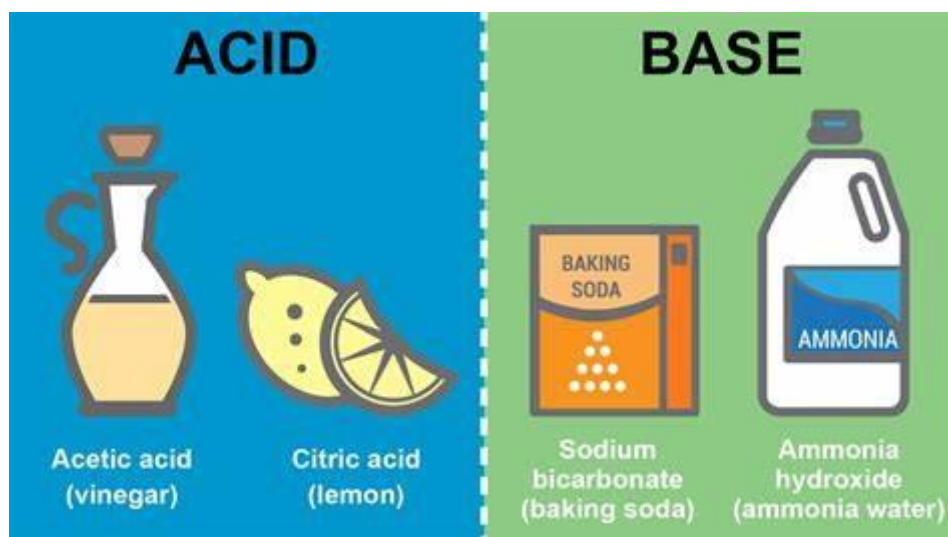
**YEAR** \_\_\_\_\_

## ACIDS AND BASES

### 2<sup>ND</sup> TERM SIMPLIFIED

**QUESTION: DO YOU UNDERSTAND THE ABOVE TOPIC?**

**READ THE FOLLOWING PICTURES**

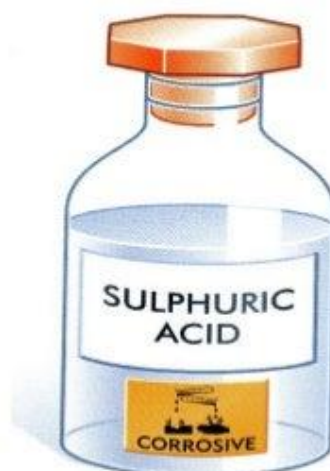


### DEFINATION ACID BASES



## Acids

- Acids are 'sour' substances
- Acids are found in fruit like lemons
- Laboratory acids like hydrochloric acid or sulphuric acid are very dangerous (corrosive)



# Main Differences Between Acid and Base

## Acid

## Base

The acid is a chemical compound that dissolves in water to give  $H^+$  ion



The acid is a chemical compound that dissolves in water to give  $H^+$  ion

Acid is a proton donor



The base is a proton acceptor

Acids have a pH value of less than 7



Bases have a pH value of more than 7

An acid turns blue litmus paper to red



Bases turn red litmus paper to blue

It can be of solid/liquid/gaseous state as per the temperature



Bases have solid and slippery state, Exception: Ammonia is present in the gaseous state

Examples of acids are acetic acid, sulphuric acid, etc



Examples of base are hydroxide, ammonia, etc.

## ACTIVITY

**GIVE 5 EXAMPLES TO SHOW UNDERSTANDING OF THE TWO CONCEPTS (10)**

| ACID | BASES |
|------|-------|
| 1.   | 1.    |
| 2.   | 2.    |
| 3.   | 3.    |
| 4.   | 4.    |
| 5.   | 5.    |

## MORE INFORMATION BASED ON ACID AND BASES

**TASTE**

| Acids                                                                                                                                                                                                                                        | Bases                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sour</b> <ul style="list-style-type: none"><li>household vinegar (acetic acid)</li><li>lemon juice, orange juice or citrus fruit juices (citric acid)</li><li>Vitamin C (ascorbic acid)</li><li>Soda (contains phosphoric acid)</li></ul> | <b>Bitter</b> <ul style="list-style-type: none"><li>Milk of Magnesia (magnesium hydroxide)</li><li>Maalox (Magnesium hydroxide and aluminum hydroxide)</li><li>Baking soda</li></ul> |



## WHAT IS THEIR USES

# USES OF BASES



**Sodium Hydroxide –**  
Soaps



**Sodium Carbonate –**  
Detergents



**Sodium bicarbonate –**  
Baking soda



**Calcium hydroxide –**  
Cement



**Ammonia –**  
Fertilizers



**Magnesium hydroxide–**  
Treating acidity



**Sodium Fluoride –**  
Toothpaste



**Bases used in –**  
Labs

# Uses of Bases

- Sodium hydroxide is in soap, paper, oven cleaners, and to unclog drains.
- Calcium hydroxide is used to make cement and plaster.
- Magnesium and aluminum hydroxides are used as antacids to treat upset stomachs.



## USES OF ACIDS

**Acids have various uses in different fields<sup>123</sup>.** Some of the uses of acids are:

- Vinegar, a diluted acetic acid solution is used as a food preservative<sup>1</sup>.
- Sulfuric acid is widely used in batteries<sup>12</sup>.
- Nitric acid and sulfuric acid are used in the industrial production of explosives, dyes, fertilizers, and paints<sup>12</sup>.
- Phosphoric acid is the main constituent in different soft drinks<sup>1</sup>.
- Hydrochloric acid is one of the components of gastric juices in the stomach. It helps in breaking down complex molecules into simpler ones that are easier to absorption<sup>3</sup>.
- Nucleic acids like RNA and DNA make up the genetic material in humans<sup>3</sup>.
- Fatty acids are the basic units of fats in our bodies<sup>3</sup>.
- Amino acids are the basic unit of proteins in our bodies<sup>3</sup>.

Learn more:

1

What are the uses of acids and bases? - BYJ...



byjus.com

## SUMMARY



### Uses of acids



#### 1 ***Preserving food***

- eg. ethanoic acid, benzoic acid, citric acid

#### 2 ***Manufacture of products like paint, detergent and fertilizers.***

- eg. sulfuric acid, hydrochloric acid and nitric acid.

#### 3 ***Batteries for vehicles***

- eg. sulfuric acid

#### 4 ***removing rust from iron or steel***

- eg. sulfuric acid

#### 5 ***making rubber from latex***

- eg methanoic acid

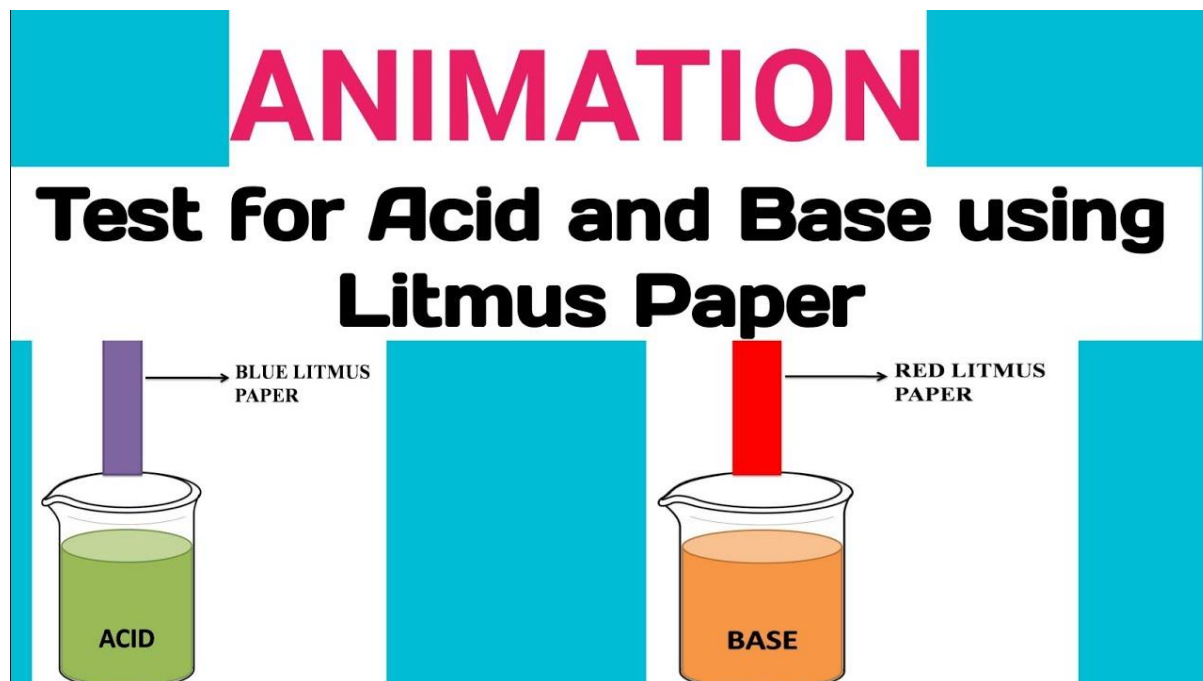
#### 6 ***Pure fruit juice***

- eg. citric acid and tartaric acid

## ACTIVITY LEARNERS HAS TO EXPLAIN HOW DO WE USE (20)

| HOW DO WE USE ACID | HOW DO WE USE BASES |
|--------------------|---------------------|
|                    |                     |
|                    |                     |
|                    |                     |
|                    |                     |
|                    |                     |

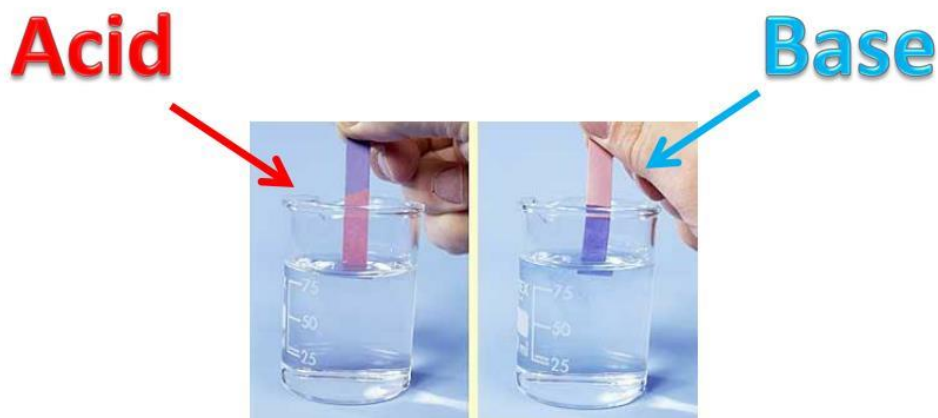
**TESTING VARIOUS SUBSTANCE WITH BLUE AND RED LITMUSPAPER**



**LEARNERS TO DRAW AND FOLLOW THE ABOVE INSTRUCTION BUT THEY MUST USE COLLORPENS**

# Reaction to litmus paper

- Can identify if a substance is acidic or basic.

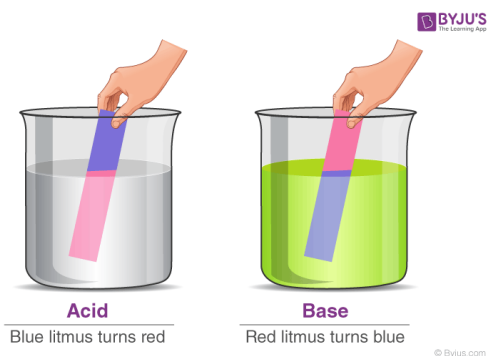


**What is your observation choose between the two**

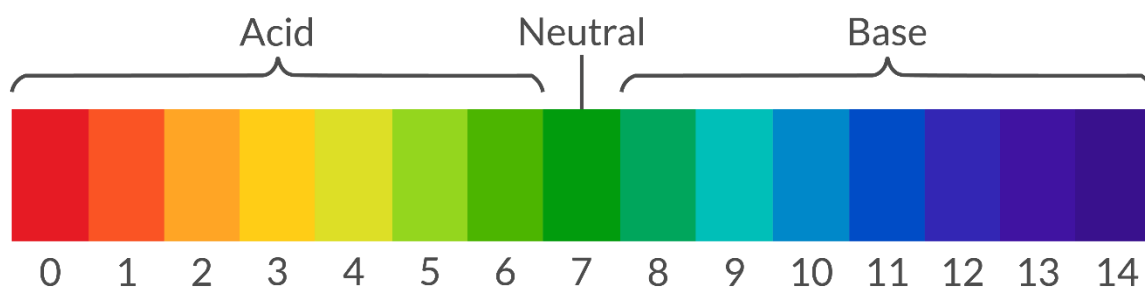
**1. On your acid glass what is your observation the blue litmus paper exposed to acid glass what is the colour\_\_\_\_\_2**

**2. On your base glass what is your observation when the red litmus paper is exposed to based solution\_\_\_\_\_2**

**WE SHOULD HAVE THE FOLLOWING RESULTS**



## SORTING ACIDS,BASES AND NEUTRALS

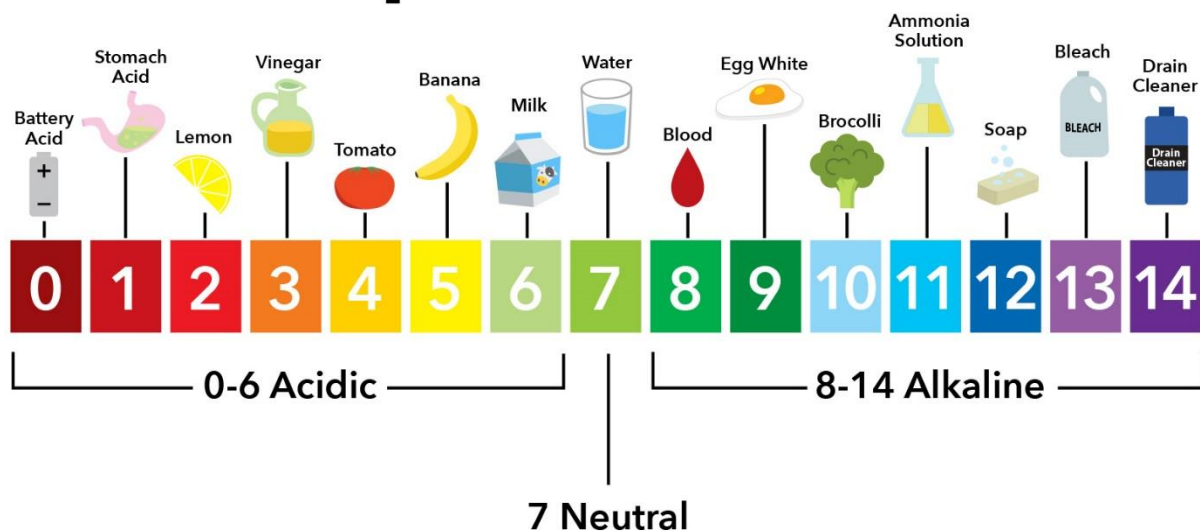


## The pH Scale

© ReAgent

**THIS WILL ASSIST WHEN YOU DEALLING WITH THIS TABLE**

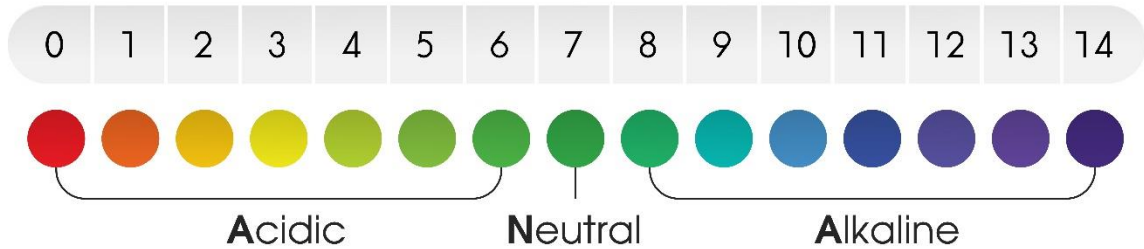
## pH Scale



**LEARNERS HAS TO SORT THE ABOVE INTO DIFFERENT COLOR  
:COLOR IN (3)**

**MAY LEARNERS FILL IN THE BELOW DIAGRAM WITH EXAMPLES OF  
ACIDS, BASES AND NEUTRALS**

# pH Scale



| ACIDS | NEWTRAL   | ALKALINE (15) |
|-------|-----------|---------------|
|       | 3 EXAMPLE |               |
|       |           |               |
|       |           |               |
|       |           |               |
|       |           |               |
|       |           |               |
|       |           |               |

## GRAND TOTAL WEEK1-3

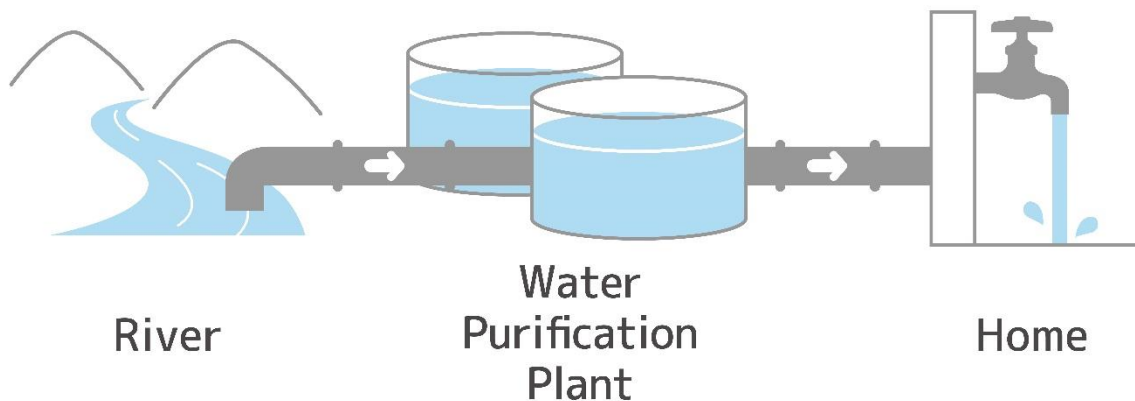
|           |
|-----------|
|           |
| (60)MARKS |

## WEEK 4-6

## PROCESS OF PURIFYING WATER

### NATURAL PROCESS

### PROCESS OF PURIFYING WATER



## ACTIVITY

**LEARNERS CAN EXPLAIN THE PROCES OF HOW DO WE GET CLEAN WATER**

**FROM 3 SOURCES**

| S 1 | S2 | S3 |
|-----|----|----|
|     |    |    |

(6)

**IMPORANCE OF DRINKING CLEAN WATER**



# Clean Water

1. Helps you control your whole body temperature
2. Assists your organism eliminate waste material
3. Boosts your mind activity, as it signifies 75% of the brain
4. Aids your cells acquire all the nutrients and also oxygen which they need for a proper perform
5. Help your body transform ingested foodstuffs in to energy
5. Works as safeguards of your current crucial body organs and cushions these as well as the important joints
6. Help your body soak up nutrients



## THESE ARE THE METHODS OF PURIFYING WATER

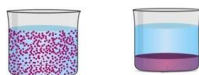


## Water Purification Methods

### Boiling



### Sedimentation



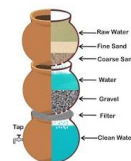
### Chlorination



### Decantation



### Filtration



## **ACTIVITY**

**LEARNERS TO FIND OUT WHAT WILL HAPPEN IF WE DRINK  
CONTAMINATED WATER MEANING WATER THAT IS NOT CLEAN OR  
PURIFIED (GOOGLE) OWN ANSWERS**

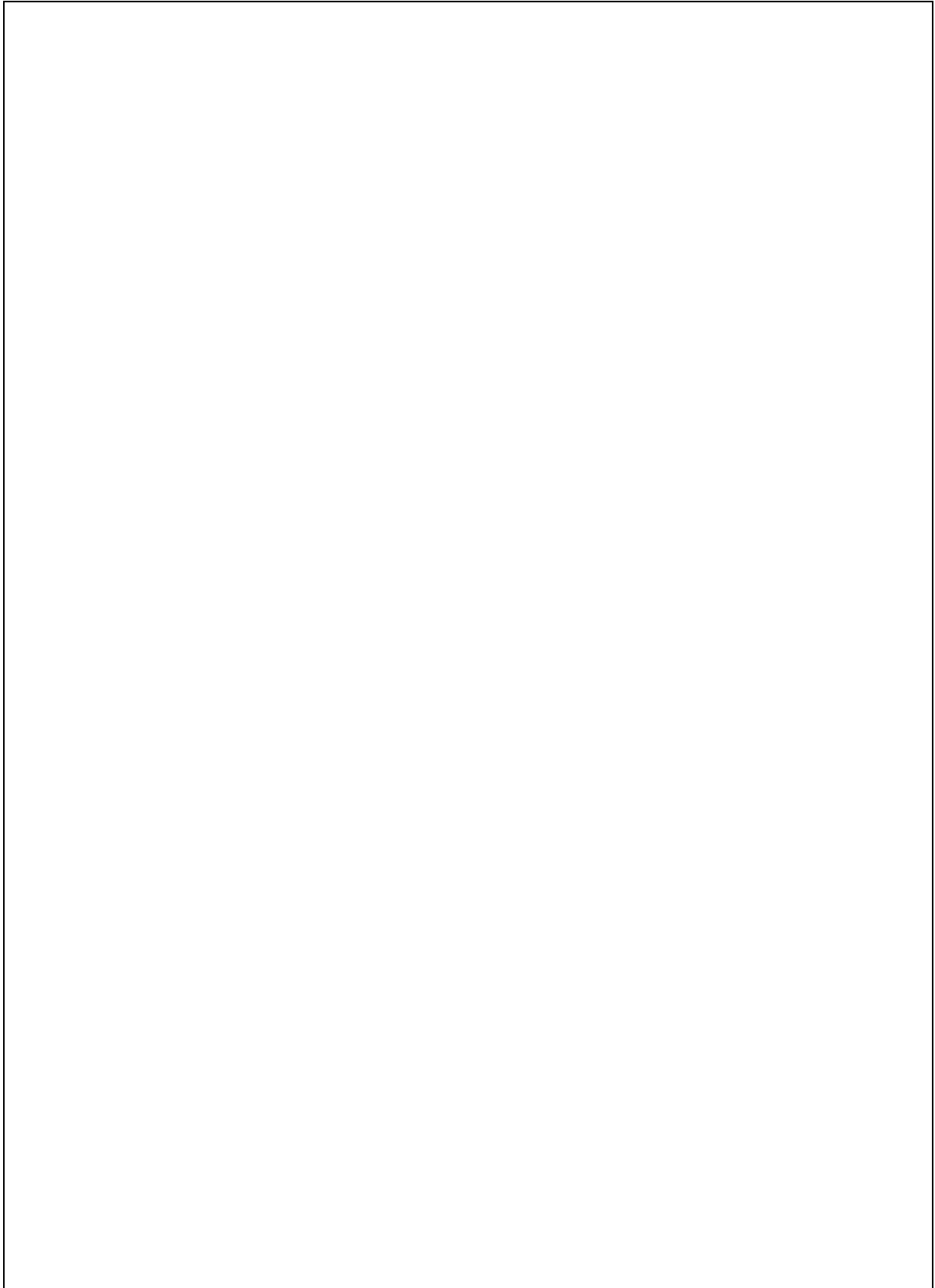
|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**(10)**

**WHY IS IMPOTANT TO DRINK CLEAN WATER (10) PASTE ANSWERS**

|  |
|--|
|  |
|--|

**DO YOU KNOW WAYS TO PURIFY WATER (15) PASTE AND DRAW**

A large, empty rectangular box with a thin black border, intended for a drawing. It occupies the majority of the page below the title.

**WHY IS IT IMPORTANT FOR HUMANS AND ANIMALS TO HAVE  
CLEAN WATER ANSWERS AFTER THE CONTENT PLS**

# Water

Animals need water to live and grow.



## Water is important

As a habitat for wild plants and animals



To transport things

For people to use (like to drink and grow food) and to get food from (like fish)



For recreation



## WATER

- All living things need water to live (plants & animals)
- Water transports nutrients to other areas
- Some organisms live in water



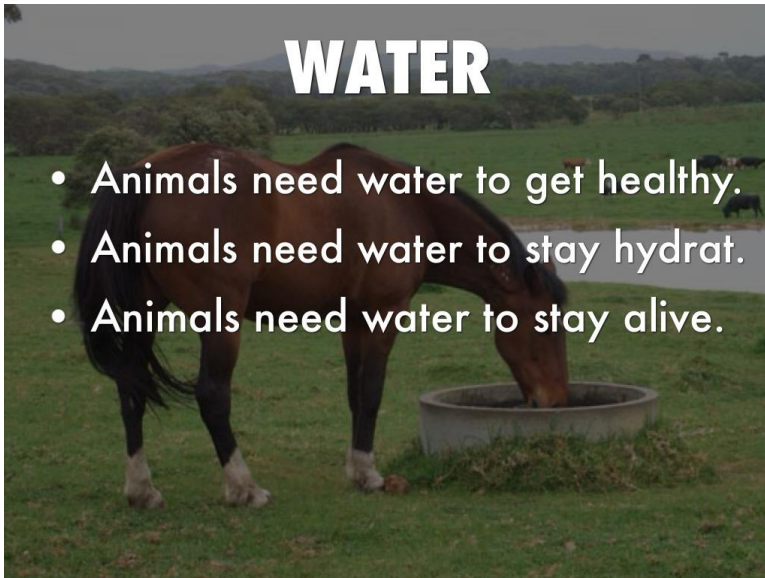
## Animals need water

- All animals need water to survive.
- Some animals live in the water.
- Some animals need a lot of water to survive.
- Some animals can survive without water for long periods of time.
- Animals intake water through drinking.



## WATER

- Animals need water to get healthy.
- Animals need water to stay hydrated.
- Animals need water to stay alive.



**COMPLETETHE TABLE JUST INDICATE WHY HUMANS,PLANTS AND ANIMAL NEED WATER (15)**

| HUMANS | PLANTS | ANIMALS |
|--------|--------|---------|
| 1.     |        |         |
| 2.     |        |         |
| 3.     |        |         |
| 4.     |        |         |
| 5.     |        |         |

## **2.2 WATER POLLUTION**

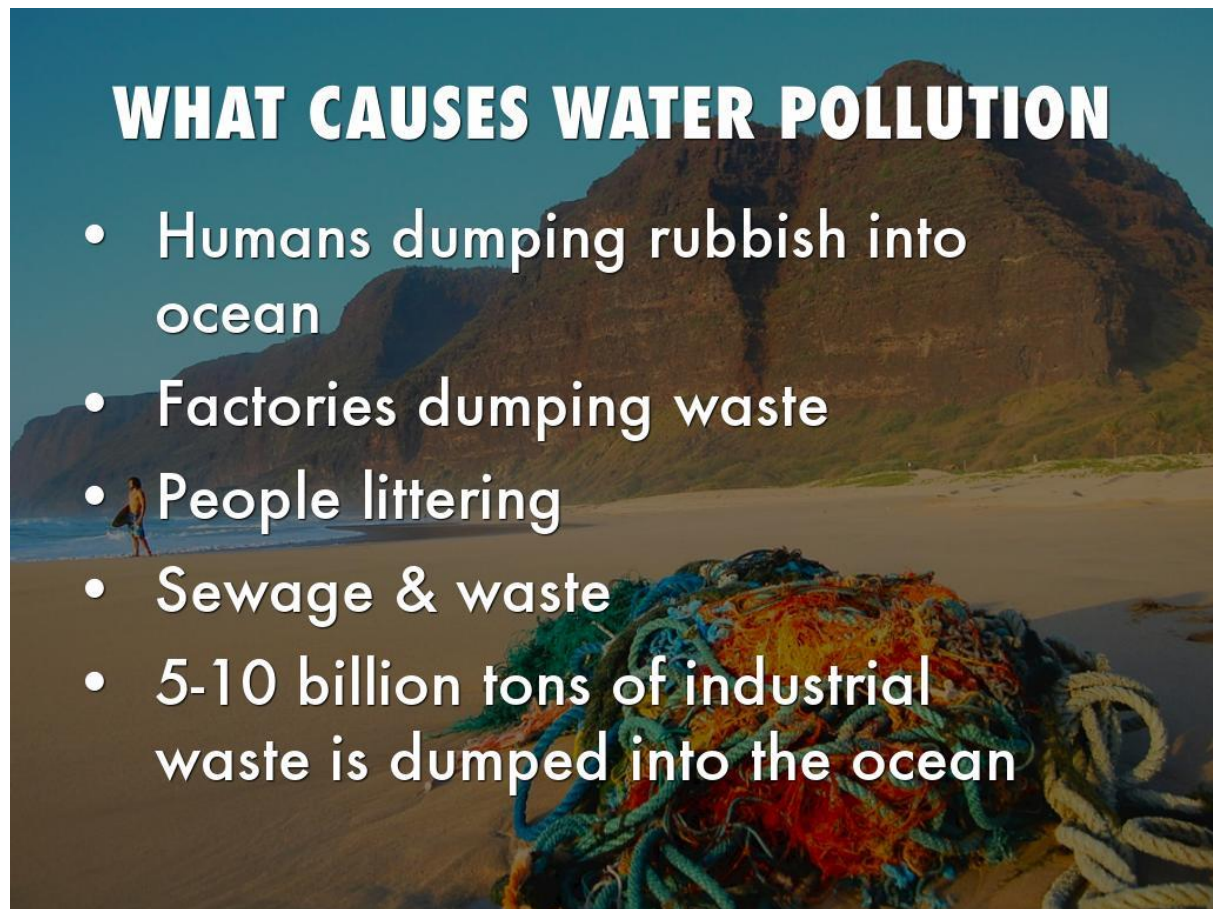
**WHAT IS WATER POLLUTION ?**

### What is water pollution?

- Water pollution is when humans dump some foreigner substances into the water, making it dirty and undrinkable. Then it is clarified as polluted. Polluted water is usually poisonous to both human and animal, making it unlivable and undrinkable.



## WHAT CAUSES WATER POLLUTION



### WHAT CAUSES WATER POLLUTION

- Humans dumping rubbish into ocean
- Factories dumping waste
- People littering
- Sewage & waste
- 5-10 billion tons of industrial waste is dumped into the ocean

**ACTIVITY FIND OUT WHAT IS POLLUTING YOUR WATER AROUND YOUR LOCAL AREA OR HAVE YOU SEEN POLLUTED WATER ANY WHERE IT MIGHT BE ON TV EXPLAIN (10)**

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |

|                                   |
|-----------------------------------|
|                                   |
|                                   |
| <b>FIND SOME POLLUTANTS PHOTO</b> |

**(5)**

**What is the importance of water for humans, animals, and plants?** Solution Importance of water: Water regulates the body temperature by the process of evaporation, sweating, and transpiration. Water is important for the germination of seeds, and the growth of plants. Water helps in the absorption of nutrients from the soil.

**What is the importance of water for humans, animals, and plants?**

...



[byjus.com/question-answer/what-](https://byjus.com/question-answer/what-is-the-importance-of-water-for-humans-animals-and-plants/)



## HOW WE USE WATER

### WORD SEARCH PUZZLE

BAKING  
BATH  
BOATING  
BRUSHING TEETH  
CLEANING  
COOKING  
DIVING  
DRINKING  
FLUSHING  
HOSE  
POOL  
RINSING  
SHOWER  
SPRINKLER  
SWIMMING  
WASHING  
WATERING

S P R I N K L E R H F I R  
C R O X W A T E R I N G W  
G U U O X H I G V Y Y G U  
I N C G L G N I K A B X O  
D I I G N I M M I W S S S  
B R U S H I N G T E E T H  
S C I S N H N G H G B B O  
O U N T I N A N O O C W  
F L U A K I R I E A S T E  
F E B D V I K U T L W E R  
A X B I V O N I I D C Y D  
Y Q D M O U N G Y R X E O  
U I Q C Y G W A S H I N G

The words appear UP, DOWN, SIDEWAYS, BACKWARDS, and DIAGONALLY.  
Find and circle each word!



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www.puzzleplay.com

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Water as a source

W B I D G V S P S X M X Y G H A V  
L G E J G Y Z E X W E R S W E E U  
P R E C I P I T A T I O N M L V O  
V F D N U I W O H H U X L Y O A H  
Y S X C X S A L T W A T E R H P N  
R Q N O I T A S N E D N O C E O U  
Z E H L Y B P B V O K O F G R R T  
R G T I P C E X D L H D D J O A W  
I I H A S W Z S H G R C I X B T D  
G M O D W Z R E V I R E F W L I G  
P H Y V B H L K E S S F Y A M O F  
X I K H R K S R V Z N I C T O N A  
G B X L F E L E J J A G W E L L H  
X L I U G H S Q R Y E M B R N P W  
J C T V S T Q E H F C F Y G V E A  
H R C C P S X C R S O L J O E N A  
G C F R E S O U R C E S K X C D J

borehole  
fresh water  
reservoir  
salt water

condensation  
oceans  
resources  
water

evaporation  
precipitation  
river  
well

Name: \_\_\_\_\_

# Water pollution

S H K X B D Y O K O F Q O P F V W R U S Z X W B  
L S X K C U N U N S G R Z N G N X S S T H T H Y  
A H N C A R E N U D Q W R V V N H T P N X C X X  
C S V J E K D X V C L S P J E P C R B A Q C P R  
I W N T A F P K T O O Y M Z X V O E N T L I E E  
M P A H Q R P Y I N D C T I I Z R A W U K X M T  
E W V I Q P F T L T J P L I W N D M K L T X Y A  
H F F I N G R C J A G F F O V O Z S H L S J N W  
C Y P L F S P I Q M G K L G E I O Y F O B I I D  
L U L M S E D I C I T S E P W T T N O P M W R N  
E N T L F A N Q T N E C D H M U J C B K U J I U  
G N J O E S M M G A F M Y R M L H P A V D T R O  
W O P U X T Y N O T C N P J Y L W R O O I F A R  
C S M A H I G O L I B O T I G O P O N X I A B G  
O W L A C W C Z I O G R X I V P N L D B G D T Y  
X Z E G P E N O F N Z V W K V L E C A J H G A U  
B U Z G E S G B T D X H A C T A X G L S R Z H R  
R F O G A T Y A R E R O O W R M P A M I T B R R  
Y W O A F B H U W N E U R U H R O I L S P I L L  
U E M M S U R J N E P N G D I E F J D S O D C K  
Y R X C A Y U A X X S Y D N F H X F Q O N E Y T  
V X H A U P D M G I M O N U U T T H P U P X J V  
W D L O M S T W V X A V I B Z P K J G U A P O C  
Y G K O W T C O V D I U H B V W K R N K J D E P

Thermal pollution

Ground water

Chemicals

Streams

Toxic

Radioactivity

Pollutants

Oil spill

Garbage

Water

Contamination

Pesticides

Plastic

Sewage

Name: \_\_\_\_\_ Date: \_\_\_\_\_

## Sources of Water Pollution

A N U N U C A J K B R M F W C Y P N X B L R A W  
B O P I J V F R N A W S I B B J U L V Y B D J J  
U Z E D I C I T S E P E B H J T S C J O W U H M  
R E C I R X M L M O E O L F U G A X B Z C N D X  
N H L O L Y O D J L E T V F A E R Y V C T K W R  
I T R A S H K E K A J X L B K D X D I P N K M O  
N K Q E C C K T P O E W C L J W U W U E M X X N  
G S T M J H E F Z C R I U A Q X N H G C C N E R  
F K S Q H Q N H M T T W U D I B S A M G D Q W O  
O K R R K I C P L S S T B R X L W V W S D S U B  
S Y M W A V W K A G V W K Y E E B M Y A B B R N  
S I J K Q C D L W J S M C Z S Z T W C W V F X P  
I Z B Q W Y P E W X Y K Q S B I I G H Y G P Z B  
L E T U K P D G I N D U S T R I A L W A S T E A  
F E I V G W F H F Z C R R Q R J W I I Z A E T D  
U F N I W X G P L E Y T Z N E P J V D T A N L X  
E J Z E O C A W F V L Y Z N G B X U E H R J P D  
L C I D L A N D F I L L L E A K S C A K E E V N  
S O F N D U L G N I N I M D O O G M H Q E G F F  
M H V N Q M C Z B Y C D I P F Q O V S H K N P V  
D N T U V O L L I P S L I O C L J R D Q L P X Q  
Y Q O D A O W M V X B R M I F U R Q P T W D T C  
Q F T T G Q M K C U Q M J T P N F H R J T B W H  
O M W V Z X Z Z O E D M W L G S O M I B B D K P

coal

landfill leaks

sewage

oil spill

cars

pesticide

mining

plastic bags

industrial waste

fertilizer

burning fossil fuels

trash

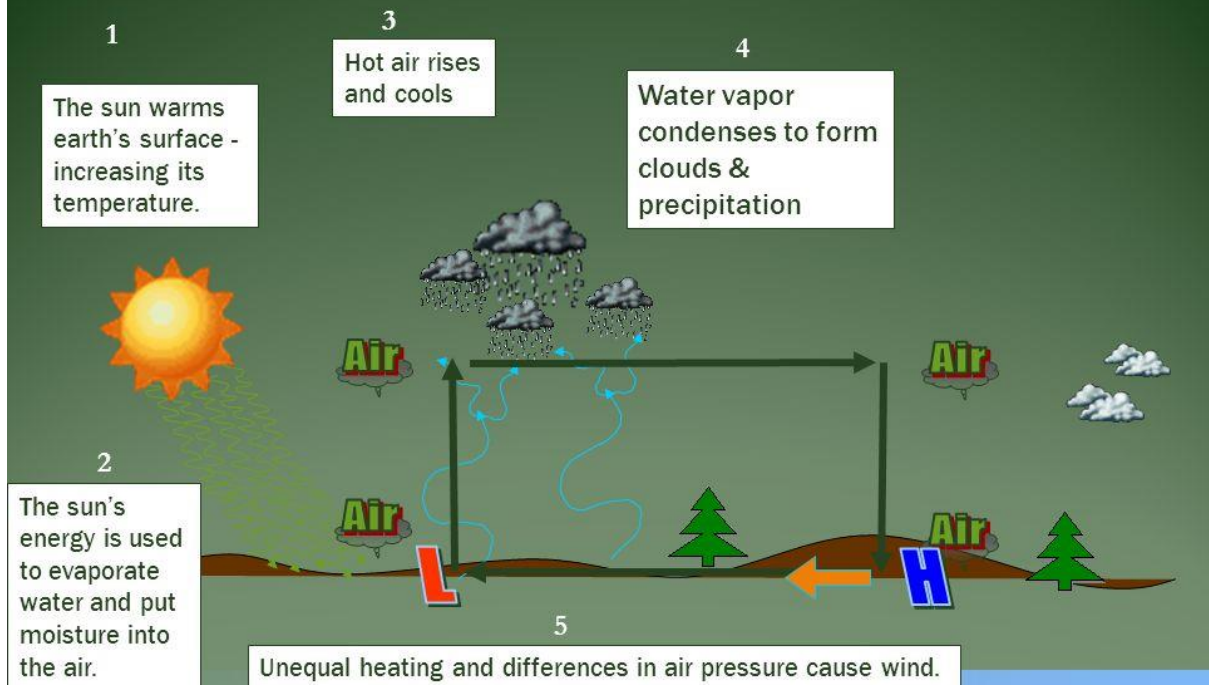
### WEEK 7-8

|              |
|--------------|
| <b>MARKS</b> |
|              |

### IMPACT OF WEATHER, TEMPERATURE AND WIND ON MATERIALS

# What causes weather?

- Caused by the transfer of energy between these 3 things:
  - The Sun, Earth's surface, and the Atmosphere.



## How Metals React in Weather

- Copper Copper is an alloy that will turn green when it is exposed to oxygen and moisture for a long time. ...
- Iron How metals react in weather especially holds true for iron and steel. ...
- Iron w/Zinc Metals react in weather of any type...and it is chemists job to try to figure out how to slow that process down. ...
- Brass ...

**Weather conditions** can significantly impact the **corrosion** of metal structures. Let's explore how different weather elements affect metals:

- Rain:**

- Rainwater, slightly acidic due to dissolved carbon dioxide, initiates a chemical reaction called **oxidation** when it contacts metal surfaces.
  - Impurities and contaminants in rain (such as airborne pollutants, salts, and minerals) act as catalysts, speeding up the corrosion process.
  - Over time, rain-induced corrosion leads to surface pitting, cracks, and fissures, compromising the metal's structural integrity.
  - Metal assets left outdoors, like bridges and railings, face prolonged exposure to rain, which can significantly reduce their expected lifespan.
  - **ZRC Cold Galvanizing Compound** is ideal for rainy climates. [It contains zinc dust, which provides corrosion protection by sacrificially corroding instead of the metal itself, safeguarding the asset from rain-induced corrosion<sup>1</sup>.](#)
2. **Temperature:**
- Generally, higher temperatures are associated with higher rates of corrosion.
  - In warmer weather, metals become more ductile (less brittle) and require more force to break.
  - [Conversely, exposure to colder temperatures makes metals more brittle and prone to breaking<sup>2</sup>.](#)
  - [You can generally expect less rust in winter and more in summer, although other factors can influence this trend<sup>3</sup>.](#)

Remember that regular **corrosion protection measures** are crucial to prevent safety hazards and costly replacements of metal assets exposed to varying weather conditions.

Learn more

1 [zrcworldwide.com](#)2 [engineeringcheatsheet.com](#)3 [hunker.com](#)

- 
- 
- 

See more

## Chemical Weathering: Rusting Metal

- This is chemical weathering because the metal on the pot mixed with a chemical and caused it to rot.



Chemical Weathering - when agents of weathering chemically change the composition of a rock.

## II. AGENTS OF CHEMICAL WEATHERING

1. Oxidation - Oxygen combines with minerals to form oxides.  
(iron + oxygen = Rust)



- Oxidation weakens the bedrock making it softer.